

ALBIN JEGUS S

AI/ML Solutions Architect| Full Stack Developer

Email: jegusselvaraj@gmail.com

Phone: +91 9750750519

GitHub: [GitHub](#)

LinkedIn: [LinkedIn](#)

Portfolio: [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Solutions Architect and Senior Full Stack Developer with 10+ years of technical experience. Specialized in Generative AI, Industrial AI, Predictive Maintenance, RAG-based LLM systems, SaaS platform development, and cloud-native application architecture. Proven record of delivering end-to-end AI-powered products across manufacturing, healthcare, education, and enterprise domains.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, Embedded C, MATLAB

AI/ML Technologies:

- TensorFlow, PyTorch, Keras, LangChain, YOLO, OpenCV, GAN Models, ResNet
- Transformers, BERT, LLM (LLaMA), NVIDIA CUDA, cuDNN, MLOps
- RAG (Retrieval-Augmented Generation) pipelines

Automation & Integration:

- n8n Workflow Automation
- Pabbly Chatflow and Connect
- WhatsApp Business API

Industrial AI & IoT:

- Predictive Maintenance
- Time-Series Forecasting
- OPC-UA
- PLC Data Integration

Web Technologies:

- Backend: Django, Flask, FastAPI
- Message Brokers: Kafka, RabbitMQ, Artemis MQ
- Frontend: React.js
- Mobile: React Native (Expo), Flutter

Cloud & DevOps:

- Azure Cloud Services
- Docker Containerization
- Jenkins CI/CD
- Deployment: Nginx, DigitalOcean, Bluehost, etc.

Databases: MySQL, MongoDB, InfluxDB 3, Vector Databases

Data Analysis: Power BI, Pandas, Matplotlib, Seaborn, PySpark, Grafana

Embedded Systems: 8051, PIC, ARM, Raspberry Pi, IoT

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE DEVELOPER:

Kaira Technology | Kovilpatti | March 2026 – Present

Architecting and delivering enterprise-grade web applications, business automation solutions, and SaaS platforms.

Key Projects:

- **Project Management System (PMS):** Developed a comprehensive SaaS-based Project Management System featuring employee attendance, task management, KPI tracking, payroll management, announcements, reporting dashboards, leave management, notifications, document management, and role-based access control.
- **Digital VCard Platform:** Developed a digital business card platform with dynamic profile management, custom card themes, QR code generation, contact sharing, analytics, payment integration, and subscription management features
- **WhatsApp and Workflow Automation:** Developed automation workflows using WhatsApp Business API, Meta Cloud API, n8n, and Pabbly Connect by integrating CRM systems, Google services, APIs, and databases to streamline business operations and customer engagement.
- **AI Voice Agent Development:** Researched and developing AI-powered voice assistant solutions with speech recognition, conversational AI, telephony integrations, and real-time response processing for customer interaction use cases

SENIOR SOFTWARE DEVELOPER:

Zhilon Solutions India Pvt.Ltd (Baytech Innovations) | Karungal | August 2023 – Feb-2026

Architected and delivered full-stack enterprise applications with optimized database design, RESTful APIs, and secure Azure cloud deployments.

Key Projects:

- **AI-Based Industrial Predictive Maintenance Platform:** Designed and developed an end-to-end Industrial IoT and AI-driven Predictive Maintenance System for real-time monitoring and failure prediction across multiple PLC-controlled manufacturing machines. Implemented OPC-UA based edge data acquisition to collect machine data and Built a scalable real-time data pipeline using Apache Kafka for streaming sensor data and InfluxDB for high-performance time-series storage. Used Grafana dashboards for live monitoring, trend analysis, KPI tracking, and alert management. Engineered LSTM-based deep learning models to predict equipment failures, detect anomalies, forecast temperature deviations, and identify maintenance requirements before machine breakdowns occurred.
- **Job Portal Website:** Developed and deployed comprehensive job portal on Azure platform with integrated Azure Blob Storage for secure document handling and Azure Active Directory for enterprise authentication
- **School Management System (SaaS):** Built scalable ERP solution for complete school operations including enrolment, staff management, and academic tracking with comprehensive reporting capabilities
- **E-commerce Platform:** Engineered full-featured multi-vendor shopping system with secure payment processing and modular architecture for scalability-(Django-MySQL)
- **Flutter-based Attendance System:** Created location-aware attendance tracking application with live GPS tracking and geofencing capabilities for enhanced security
- **Face Recognition System:** Designed and deployed YOLO-based real-time facial recognition module for secure attendance and access control with integrated admin

dashboard

- **Medical Recommendation System:** Developed advanced AI model using Llama-2-7b-hf with custom dataset implementation using RAG logic and MLOps pipeline

SOFTWARE ENGINEER A3:

Partnersoft Technology | Chennai | October 2022 - July 2023

Led end-to-end project development from scratch using agile methodologies. Created detailed architecture and flow diagrams using Gliffy. Specialized in AI/ML integrations including AR, NLP, and OCR technologies.

Key Projects:

- **Virtual Try-On System(3D):** Implemented AR-based virtual clothing simulation platform enabling users to try outfits in real-time using advanced image and depth processing algorithms
- **OCR & Document Digitization:** Built custom OCR pipeline enhanced with NLP capabilities for extracting structured data from scanned documents with high accuracy

PYTHON PROGRAMMER:

Mentor Thesis | Marthandam | November 2021 - October 2022

Developed deep learning solutions across healthcare, surveillance, and security domains. Specialized in implementing serverless cloud-based ML workflows using AWS Lambda. Focused on real-time systems, intelligent video processing, and biometric authentication technologies.

Key Projects:

- **Breast Cancer Detection:** Developed deep learning model using MRI scans, achieving 98% accuracy for early-stage breast cancer classification
- **Human Action Recognition:** Created video analytics system that recognized human activities and generated real-time subtitles using computer vision techniques
- **Server less ML Deployment:** Deployed ML models using AWS Lambda and API Gateway, ensuring scalability and reduced infrastructure costs
- **License Plate Detection:** Built real-time vehicle license plate recognition system optimized for traffic surveillance environments
- **Face Authentication System:** Developed biometric authentication system using Hierarchical Attentional Model for high-accuracy face verification

SOFTWARE TECHNICAL TRAINEE:

IBM (Adroit Technologies) | Coimbatore | September 2020 - October 2021

Engineered practical ML solutions and analytics projects for retail and customer behavior insights using Python and TensorFlow.

Key Projects:

- **Retail Purchase Prediction:** Developed ML models using TensorFlow to forecast customer buying behavior and optimize inventory strategies for retail clients
- **Customer Behavior Analysis:** Conducted advanced data analysis using Pandas,

NumPy, and visualization libraries to uncover actionable trends in large-scale customer datasets

- **Corporate Training Delivery:** Facilitated structured sessions for working professionals and final-year engineering students in AI, ML workflows, data pipelines, and real-world case studies

EMBEDDED TRAINER:

Xplore IT Corp | Coimbatore | January 2020 - September 2020

Conducted technical training programs focused on embedded systems and real-time microcontroller applications. Specialized in IoT-based data systems using PIC and ARM controllers.

Key Projects:

- **IoT-Based LOOM Monitoring System:** Designed and implemented real-time monitoring solution using PIC microcontroller for data acquisition and analytics in textile manufacturing environments

ASSISTANT PROFESSOR:

Park College of Engineering and Technology | Coimbatore | April 2014 - December 2019

- Taught undergraduate courses in Electronics and Communication Engineering
- Supervised student projects and research initiatives in embedded systems and signal processing

EDUCATION

- **Master of Engineering- Communication System** - PET engineering college
- **Bachelor of Engineering - Electronics and Communication Engineering**
VINS Christian College of Engineering

RESEARCH & INNOVATIONS

Advanced AI Models Implemented:

- Capsule Triple GAN, Hierarchical Attention Neural Networks
- Tree CNN, DeTraC Deep CNN, Capsule Super Resolution GAN
- Automatic Graph Neural Networks, Enhanced Capsule GAN
- Recalling Enhanced RNN

Optimization Algorithms:

- Rat Swarm Optimization
- Red Fox Algorithm
- Nomadic People Optimizer
- Tunicate Swarm Algorithm